

$$\begin{aligned} & \frac{1}{m_e^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} + \frac{1}{16\pi^2} \left(\frac{1}{72} \frac{1}{m_b^2} s_\gamma^2 \left(\overline{y}_d^{i4i3} \left(27 \overline{y}_e^{\text{pr}} y_e^{pi2} y_e^{i1r} \left(1 + s_\gamma^2 \right) + 2 y_e^{i1i2} \right. \right. \right. \right. \\ & \quad \left. \left. \left. \left(19 g_1^2 + 81 g_2^2 - 24 g_3^2 - 18 c_\gamma^2 \left(3 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} + \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \right) \right) \right) + \right. \right. \\ & \quad \left. \left. 9 y_e^{i1i2} \left(3 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} \left(1 + s_\gamma^2 \right) - \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{\text{pr}} \left(-1 + c_\gamma^2 \right) \right) \right) \right) + \\ & \frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \left(2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma} \right) \text{LF}_{1,0} \left[m_d^{\text{P}} \right] + \\ & 3 s_\gamma \frac{1}{m_b^4} \overline{y}_d^{\text{pr}} \overline{y}_d^{i4i3} y_d^{\text{pr}} y_e^{i1i2} \left(-s_{2\gamma} c_\gamma + s_\gamma^3 \right) \text{LF}_{1,0} \left[m_d^{\text{r}} \right] + \\ & \frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \left(2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma} \right) \text{LF}_{1,0} \left[m_e^{\text{P}} \right] + \\ & s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} \overline{y}_e^{\text{pr}} y_e^{\text{pr}} y_e^{i1i2} \left(-s_{2\gamma} c_\gamma + s_\gamma^3 \right) \text{LF}_{1,0} \left[m_e^{\text{r}} \right] + \frac{1}{2} s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\ & \quad \left(2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \left(-s_{2\gamma} c_\gamma + s_\gamma^3 \right) - \sum_p g_1^2 \left(2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma} \right) \right) \text{LF}_{1,0} \left[m_l^{\text{P}} \right] + \frac{1}{2} s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\ & \quad \left(6 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \left(-s_{2\gamma} c_\gamma + s_\gamma^3 \right) + 6 c_\gamma \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \left(s_{2\gamma} + s_\gamma c_\gamma \right) + \sum_p g_1^2 \left(2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma} \right) \right) \\ & \quad \text{LF}_{1,0} \left[m_q^{\text{P}} \right] - \sum_p s_\gamma g_1^2 \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \left(2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma} \right) \text{LF}_{1,0} \left[m_u^{\text{P}} \right] + \\ & 3 s_\gamma c_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \left(s_{2\gamma} + s_\gamma c_\gamma \right) \text{LF}_{1,0} \left[m_u^{\text{r}} \right] + \frac{1}{4} s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\ & \quad \left(3 s_{4\gamma} c_\gamma \left(g_1^2 + g_2^2 \right) + s_\gamma \left(g_1^2 \left(-1 + 3 c_{2\gamma}^2 \right) + 3 g_2^2 \left(-1 + c_{2\gamma}^2 \right) \right) \right) \text{LF}_{1,0} \left[m_\Phi \right] + \\ & \frac{1}{4} \frac{1}{m_b^2} s_\gamma^2 \left(\overline{y}_d^{i4i3} \left(-3 s_\gamma^2 \overline{y}_e^{\text{pr}} y_e^{pi2} y_e^{i1r} + 2 y_e^{i1i2} \left(g_1^2 + 3 g_2^2 \right) \right) + \right. \\ & \quad \left. 3 y_e^{i1i2} \left(-s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} + c_\gamma^2 \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{\text{pr}} \right) \right) \text{LF}_{1,1} \left[m_\Phi \right] - \\ & \frac{1}{4} s_\gamma^2 y_e^{i1i2} \left(\overline{y}_d^{i4i3} \left(g_1^2 + 3 g_2^2 \right) + 2 c_\gamma^2 \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{\text{pr}} \right) \text{LF}_{1,2} \left[m_\Phi \right] - \\ & \frac{1}{9} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_1, m_d^{i3} \right] + \frac{1}{18} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_1, m_d^{i3} \right] - \\ & g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_1, m_e^{i2} \right] + \frac{1}{2} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_1, m_e^{i2} \right] - \\ & \frac{1}{4} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_1, m_l^{i1} \right] + \frac{1}{8} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_1, m_l^{i1} \right] - \\ & \frac{1}{36} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_1, m_q^{i4} \right] + \\ & \frac{1}{72} g_1^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_1, m_q^{i4} \right] - g_1^2 \frac{1}{m_b^4} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,-1} \left[m_1, \tilde{\mu} \right] - \\ & 2 m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \left(c_\gamma^2 - 2 s_\gamma^2 \right) \text{LF}_{1,1,0} \left[m_1, \tilde{\mu} \right] - \\ & \frac{3}{4} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_2, m_l^{i1} \right] + \\ & \frac{3}{8} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_2, m_l^{i1} \right] - \frac{3}{4} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_2, m_q^{i4} \right] + \\ & \frac{3}{8} g_2^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_2, m_q^{i4} \right] - 3 g_2^2 \frac{1}{m_b^4} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,-1} \left[m_2, \tilde{\mu} \right] - \\ & 6 m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \left(c_\gamma^2 - 2 s_\gamma^2 \right) \text{LF}_{1,1,0} \left[m_2, \tilde{\mu} \right] - \\ & \frac{4}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_3, m_d^{i3} \right] + \frac{2}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_3, m_d^{i3} \right] - \\ & \frac{4}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_3, m_q^{i4} \right] + \frac{2}{3} g_3^2 \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} \text{LF}_{2,1,-1} \left[m_3, m_q^{i4} \right] - \\ & \frac{1}{2} \frac{1}{m_b^2} s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} y_e^{i1i2} \text{LF}_{1,1,0} \left[m_d^{\text{P}}, \tilde{\mu} \right] + 3 s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\ & \quad \left(s_\gamma \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \left(-2 c_\gamma^2 + s_\gamma^2 \right) - \tilde{\mu} c_\gamma \left(c_\gamma^2 - 2 s_\gamma^2 \right) \left(\overline{y}_d^{\text{pr}} a_d^{\text{pr}} + y_d^{\text{pr}} \overline{a}_d^{\text{pr}} \right) + 3 s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \right) \\ & \quad \text{LF}_{1,1,0} \left[m_d^{\text{r}}, m_q^{\text{P}} \right] + s_\gamma \frac{1}{m_b^4} \overline{y}_d^{i4i3} y_e^{i1i2} \\ & \quad \left(s_\gamma \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \left(-2 c_\gamma^2 + s_\gamma^2 \right) - \tilde{\mu} c_\gamma \left(c_\gamma^2 - 2 s_\gamma^2 \right) \left(\overline{y}_e^{\text{pr}} a_e^{\text{pr}} + y_e^{\text{pr}} \overline{a}_e^{\text{pr}} \right) + 3 s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \right) \\ & \quad \text{LF}_{1,1,0} \left[m_e^{\text{r}}, m_l^$$